

How computer programs work

Computers might seem very smart, but they are actually just boxes that follow instructions very quickly and accurately. As intelligent humans, we can get them to carry out different tasks by writing programs, or lists of instructions.

1 Computers can't think

A computer won't do anything by itself. It's up to the computer programmer to give it instructions.



Without instructions a computer is clueless

2 Write a program

You can tell a computer what to do by writing a set of very detailed instructions called a program. Each instruction has to be small enough that the computer can understand it. If the instructions are incorrect, the computer won't behave the way you want it to.

This is a computer program counting down to launch

```
for count in range(10, 0, -1):
    print("Counting down", count)
```

3 Programming languages

Computers can only follow instructions in a language they understand. It's up to the programmer to choose which language is best for the task.

```
for count in range(10, 0, -1):
    print("Counting down", count)
```

All programs are finally converted into "binary code", a basic computer language that uses only ones and zeroes

```
0010 0011 1000 1100
1000 0110 0100 1001
0100 1001 0001 0101
```

BLAST OFF!



LINGO

Hardware and software

"Hardware" means the physical parts of the computer that you can see or touch (all the wires, the circuits, the keyboard, the display screen, and so on). "Software" means the programs that run on the computer and control how it works. Software and hardware work together to make computers do useful things.